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## **AI-Enabled Predictive Maintenance Coupled with Marine-Logistics Responder for Offshore Energy Islands Infrastructure**

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### **Abstract**

Offshore energy islands host critical marine civil assets whose performance is vital to ensuring national and global energy security. In the context of climate change, these assets face growing risks of progressive degradation. Structures such as reclaimed platforms, jetties, dolphins, pontoons, navigation aids, intakes, rock protections, and quay walls, though designed for strict metocean conditions, are increasingly challenged by both foreseen and unforeseen environmental conditions. Safeguarding their integrity and ensuring long-term resilience requires moving beyond fragmented inspections toward a systematic, data-driven process. This work presents an AI-enabled data to action framework that transforms diverse monitoring streams into preventive maintenance plans, thereby enhancing resilience, reducing lifecycle costs, and supporting global sustainability goals. The framework aims at consolidating monitoring data into a unified digital environment that enables predictive modeling and dynamic risk updates. Advanced analytics and machine learning techniques identify degradation trends, quantify uncertainty, and refine risk profiles over time. A decision-support layer translates these insights into prioritized preventive actions through multi-criteria evaluation, balancing reliability, operational continuity, and sustainability objectives. Designed to be compatible with existing asset management systems, the framework provides asset owners with a structured process to strengthen and enrich current practices. By embedding monitoring, evaluation, and learning into a closed loop process, and by adding a governance layer that ensures traceability and alignment with best practices, the framework supports the transition from reactive inspection to proactive infrastructure management. In this work, the simulation results illustrate how complex datasets can be aggregated into transparent, simple maintenance strategies. The approach enables predicting deterioration trends, optimizes resource allocation, and reduces exposure to unplanned disruptions, particularly vital for energy islands with constrained operational windows. By integrating performance, cost, and sustainability criteria in one process, the framework supports balanced decision-making across technical and managerial levels. Its applicability extends naturally to ports, coastal protections, and other offshore infrastructure, making it a scalable model for systematic predictive maintenance. A key addition in this work is the explicit consideration of marine logistics in the process, which is a factor often dominating the cost and carbon footprint of offshore maintenance. By reducing unnecessary vessel deployments and optimizing intervention timing, the approach lowers both operational risks and environmental impacts, especially where islands are

located in marine protected areas. The novelty of this work lies in combining advanced data analytics with structured asset management practices into a consistent, data to intervention process, offering a practical way forward for more reliable, efficient, and sustainable infrastructure management.

## Introduction

Marine civil infrastructure underpins the operational reliability of offshore energy islands and associated facilities. These assets are increasingly exposed to progressive degradation due to harsh environmental conditions and the compounded effects of climate variability, making their sustainable stewardship both a technical and strategic priority.

Conventional practice for managing marine civil assets on offshore energy islands is largely based on scheduled inspections, periodic hydrographic and geotechnical surveys, and empirical design checks against established metocean criteria. While these approaches have been effective for ensuring compliance with design standards, they remain predominantly reactive and fragmented. Condition assessments are often limited to snapshot measurements, making it difficult to capture progressive degradation trends such as cracking, settlement, scour, or corrosion.

A further limitation lies in the operational logistics required to support inspections and repairs. Mobilizing marine vessels, diving teams, or specialist survey equipment can be expensive and carbon intensive, with interventions often planned on fixed schedules rather than evidence-based priorities. As a result, asset owners face excessive costs, exposure to unplanned failures, and challenges in aligning maintenance activities with allocated budgets and sustainability objectives.

From a methodological standpoint, available asset management frameworks in the marine and offshore domain tend to emphasize documentation and compliance rather than predictive and adaptive decision-making. Data are collected in passive ways, with minimal integration across geotechnical, structural, and environmental streams, limiting the ability to build a continuous understanding of asset condition. Recent advances in digitalization and analytics have been widely adopted in the oil and gas sector for subsurface and production optimization, yet their application to the civil infrastructure supporting offshore energy islands remains underdeveloped.

These gaps highlight the need for a systematic, data-driven methodology that not only consolidates diverse monitoring streams but also translates them into timely and clear maintenance actions. Such an approach would bridge the divide between data collection and decision-making and help operators move from reactive responses to proactive asset management of critical marine assets.

The imperative for resilient and sustainable marine civil infrastructure is reinforced by global policy objectives and recent advances in engineering standards. This work aligns with these frameworks through:

- Global policy alignment:
  - The United Nations Sustainable Development Goals (SDGs) [1] especially SDG 9 (industry, innovation & infrastructure), SDG 11 (sustainable cities & communities), SDG 13 (climate action), and SDG 14 (life below water) emphasize the importance of infrastructure that is durable, adaptive, and environmentally responsible.
  - The Organisation for Economic Co-operation and Development (OECD) [2] in its Principles on the Governance of Critical Infrastructure Resilience report [2] promote a systematic shift from reactive asset protection to forward-looking system resilience, offering a guideline that enables governments and operators to coordinate across sectors, prioritize resilience investments, and embed lifecycle risk governance in infrastructure policy.
- Professional and institutional guidance:

- The recent ISO 55000 series [3] establishes value-driven asset management principles through coherent alignment of governance, risk, and performance, providing a foundation for adaptation to marine civil contexts.
  - The Institution of Civil Engineers (ICE) in the recent Guiding Principles of Asset Management [4] advocates lifecycle-oriented asset management that integrates monitoring, performance, and decision-making in structured frameworks.
- Technical standards for marine infrastructure:
  - The Second Generation Eurocodes, particularly recent updates to Eurocode 8, introduce key performance-based enhancements, including refined limit states (e.g., Damage Limitation, Significant Damage) and explicit provisions for assessment, retrofitting, and structural robustness. This represents a clear shift from force-based to displacement-based and performance-based design evaluation. This recent change aligns closely with the previous PIANC's Seismic Design Guidelines for Port Structures [5], which pioneered the use of damage limitation categories tied to performance grades such as Serviceable, Repairable, Near Collapse, and Collapse. These categories are readily integrable within the proposed Bayesian risk evaluation framework, adopted in this work.
  - A recent PIANC EnviCom technical note on Climate Change Adaptation for Ports and Inland Waterways [6] provides strategies for climate change adaptation in port and waterway infrastructure, reinforcing the need for systematic, resilience-focused methods in marine environments.

The aim of this paper is to introduce and evaluate an AI-enabled data-to-action framework for the proactive management of marine civil infrastructure on offshore energy islands. The framework consolidates multi-source monitoring data, applies advanced analytics and Bayesian risk evaluation to identify degradation trends, and translates results into prioritized preventive maintenance actions. By embedding performance categories such as those defined in international and maritime guidelines (e.g., Serviceable, Repairable, Near Collapse, Collapse) within a probabilistic decision layer, the methodology ensures that monitoring outputs are linked to performance objectives. The paper demonstrates how this approach enables a structured transition from reactive inspection to proactive, performance-based asset management. Beyond the offshore energy island context, the framework is presented as a scalable model that can be extended to ports, coastal defenses, and other marine civil assets, aligning operational decisions with national, international standards and global sustainability goals.

## Statement of Theory and Definitions

### Risk and Resilience Foundations

**Resilience Definition.** Resilience in the context of marine civil infrastructure is the ability of assets to withstand, adapt to, and recover from progressive degradation or extreme events while maintaining essential functionality and minimizing long-term performance loss.

**Predictive Analytics.** Predictive analytics refers to the use of statistical learning and machine-/deep-learning models to estimate future states of an asset (e.g., condition, performance, degradation rate) from historical and real-time data such as loads, environment, inspections, and sensor streams.

**Monitoring Techniques.** Monitoring techniques provide the fundamental data streams required to evaluate degradation processes, quantify uncertainty, and support resilience-oriented asset management. In the context of marine civil infrastructure, they include both traditional survey methods and advanced sensing technologies. Typical approaches include:

- Remote sensing and geospatial methods (e.g., InSAR, UAV photogrammetry, LiDAR, sonar) to capture large-scale settlement, scour, and shoreline dynamics.
- Structural and material monitoring (e.g., ultrasonic testing, corrosion potential surveys, fiber-optic sensors, strain gauges) to assess deterioration at component level.
- Geotechnical and environmental monitoring (e.g., settlement plates, piezometers, metocean buoys) to track subsurface and environmental drivers of degradation.

These techniques are not standalone, in fact, their value increases when integrated into a unified digital environment that supports predictive analytics and Bayesian risk evaluation. By enabling continuous, multi-scale observation, monitoring provides the evidence base for transitioning from reactive inspections to proactive, resilience-focused asset management.

An example of different monitoring and sensing techniques is presented in [Table 1](#).

Table 1—Different monitoring and sensing techniques

| Technique                                       | Active / Passive | Location class | Location | Role                                                                                                 |
|-------------------------------------------------|------------------|----------------|----------|------------------------------------------------------------------------------------------------------|
| InSAR (spaceborne SAR)                          | Active           | Remote         | Space    | Portfolio-scale displacement rates; Line-of-Sight displacement (LOS) converted to vertical movement. |
| UAV photogrammetry                              | Passive          | Remote         | Air      | Orthos/3D models; crest profiles; fine post-event change.                                            |
| UAV LiDAR                                       | Active           | Remote         | Air      | High-fidelity 3D in low texture/glare; complements photogrammetry.                                   |
| ROV/AUV multibeam sonar                         | Active           | Proximal       | In-water | Toe/berm morphology, scour, block alignment under water.                                             |
| ROV/handheld UT thickness                       | Active           | Proximal       | Contact  | Steel pile wall-loss trends; splash/tidal zones.                                                     |
| Fibre-optic SHM (FBG/DAS)                       | Active           | In-place       | In-Place | Continuous strain/temperature/tilt; early warning.                                                   |
| CP potential surveys (Ag/AgCl)                  | Passive          | Proximal       | In-Place | Protection level, rectifier tuning; polarization decay checks.                                       |
| Settlement plates, inclinometers, extensometers | Passive/Active*  | In-place       | In-Place | Ground/structural movement; manual or datalogger readouts.                                           |
| Wave buoy / metocean                            | Active           | In-place       | Moored   | Wave/level forcing; links events to observed responses.                                              |

\* Inclinometers/extensometers are read with active instruments; plates are passive gauges.

**Probabilistic vs Deterministic Approaches.** Deterministic approaches for design rely on fixed values and safety factors, assuming that loads and material properties can be defined with certainty. While effective for initial design checks, they offer limited information into how structures behave under variable conditions or progressive degradation.

Probabilistic approaches, explicitly account for uncertainty in actions, material properties, and deterioration rates. They describe outcomes in terms of likelihoods and distributions rather than single values, providing a more realistic basis for risk-informed decision-making.

In the context of marine civil infrastructure, adopting a probabilistic framework is essential to capture the combined influence of environmental variability, material degradation, and operational constraints. This enables performance and resilience to be assessed over the asset lifecycle and supports the transition from compliance-driven checks to predictive, evidence-based asset management.

**Bayesian Updating.** Bayesian updating provides a systematic way to combine prior knowledge with new monitoring data in order to reduce uncertainty and refine predictions of asset condition. In practice, it allows

asset managers to start with an initial estimate of degradation or risk (the prior), update it as new evidence becomes available (the likelihood), and obtain an improved assessment (the posterior).

**Performance grades and limit states.** The performance grade refers to a qualitative category describing the expected condition or functionality of a structure under given actions (e.g., serviceable, repairable, near collapse, collapse). The limit state refers to the quantitative threshold beyond which a structure no longer meets a specified performance requirement, such as safety, serviceability, or durability.

In this framework, we adopt PIANC (WG34) definitions, and we define four main performance grades that describe acceptable levels of damage under external actions:

- Serviceable: Structure remains fully functional with little or no damage.
- Repairable: Damage occurs but can be repaired without loss of overall stability.
- Near Collapse: Severe damage occurs, but global failure is avoided.
- Collapse: Structural failure, with loss of functionality and safety.

In the proposed framework, these performance categories are integrated into the Bayesian risk evaluation process, ensuring that monitoring outputs are mapped to established performance objectives.

### Core Elements of the Framework

**Digital Twin.** A Digital Twin (DT) is a digital representation of the asset condition and performance. It is used to fuse geometry, materials, boundary conditions, and monitor changes to reflect the current state.

In this work, the DT provides the state input to the Bayesian decision layer. The DT assures tracking the change in the state parameters of the actual physical asset. A fully realized DT provides access to a detailed real-time coupling of geometry, physics, and boundary conditions. A DT, as an comprehensive medium, links monitoring data streams with deterministic/probabilistic degradation models, enabling forecasting of state transitions, i.e., from safe to alert to failure. It can form a the digital backbone of this proposed framework, bridging the gap in terms of predictive analytics, optimization, and logistics planning as to the actual physics and conditions of the asset.

The implementation of a full digital twin can be challenging and time-consuming, especially considering an operational portfolio of assets. In fact, AI powered web-based platforms with 3D illustration of the asset can also provide a highly practical and cost-effective step toward implanting this proposed framework for an AI-driven predictive maintenance for marine civil infrastructure. Such a platform integrates multi-source monitoring data, inspection results, and degradation indicators into a single, accessible interface. A smart platform can support in data aggregation, visualization, and decision support. This allows asset owners and operators to begin reaping the benefits of predictive maintenance with significantly lower investment and shorter lead times. The smart web-based platform can connect theory to practice from degradation models, embedding the risk manager logic, structuring action recommender workflows, and linking to the marine responder. It can offer ready predictive insights and optimized intervention strategies to asset owners.

**Degradation Model.** A Degradation Model (DM) is a model that mathematically represents how the condition of an asset deteriorates over time under given loads, environments, and usage. It translates physical, chemical, and mechanical processes into probabilistic or deterministic functions that predict state transitions. Key attributes of degradation models in the context of marine civil infrastructure include:

- Electrochemical degradation: corrosion of steel piles, reinforcement, or connectors
- Hydrodynamic degradation: armour unit displacement, scour, overtopping
- Geotechnical degradation: creep, consolidation, sink holes

- Structural degradation: fatigue, cracking, wall-loss

Mathematically, a degradation model, as a simple example, can be expressed as:

$$X(t + \Delta t) = X(t) + g(X(t), \theta, E) \times \Delta t + \epsilon \quad \text{Eq. 1}$$

Where:

$X(t)$  is a state variable (e.g., thickness, settlement, displacement).

$g(\cdot)$  is a degradation function describing the evolution rate.

$\theta$  is the material parameters (e.g., corrosion rate constant, creep coefficient).

$E$  is the environmental/external loading factor (e.g., salinity, impact, wave height, surcharge, fines content).

$\epsilon$  is the uncertainty or noise term.

In fact, the previous equation Eq. 1, suggests a linear (or multi-linear) behavior of degradation over time, however, in reality, this might appear very simplistic, and not reflect the actual state of performance over time. The reason being that several degradation factors contribute differently to the aging of the structure, therefore, in order to set a reliable model, it is worth involving subject matter expertise.

Not all states are equally tolerable. In fact, engineering design codes establish ultimate and serviceability limit states (ULS/SLS), and these can be recast into safety/efficacy gates for predictive maintenance for example, a residual capacity margin should not be below the unity as shown in Eq. 2, when partial factors are used in the design, e.g., when adopting recent BS 6349 or Eurocode standards, it is

$$\frac{R_d}{E_d} \geq 1.0 \quad \text{Eq. 2}$$

Where:

$R_d$  as design resistance

$E_d$  as design effect of action

By parameterizing degradation models, the degradation model can simulate and forecast asset deterioration trend, which then feed into the Bayesian decision network for risk-informed action selection. In fact, the digital twin reflect the current state, while the DM predicts its evolution.

**Bayesian Decision Network.** A Bayesian Decision Network (BDN) is a probabilistic graphical model that integrates condition indicators, failure states, and consequences to support risk informed decisions leading to the selection of the best preventive maintenance action of the relevant asset (Fig. 1). It extends a standard Bayesian network by incorporating decision and utility nodes, making it suitable for asset management applications where trade-offs must be evaluated systematically.

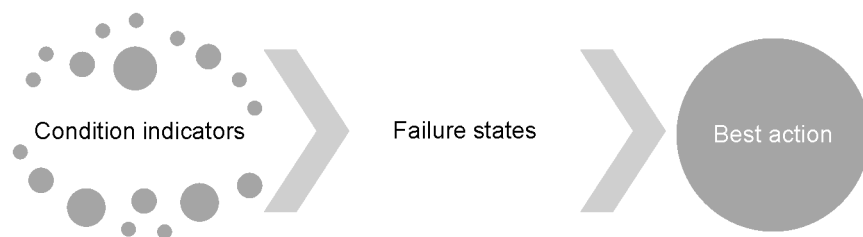
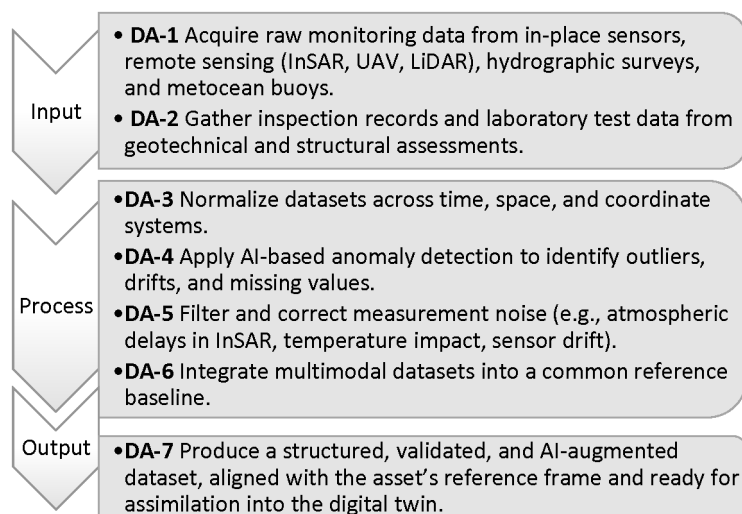
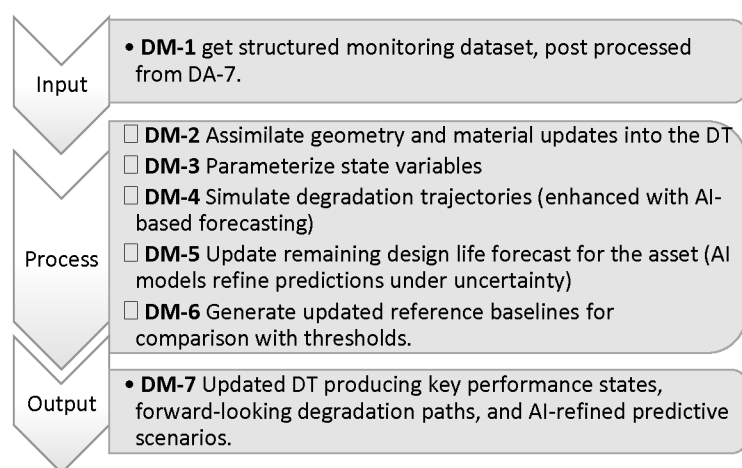


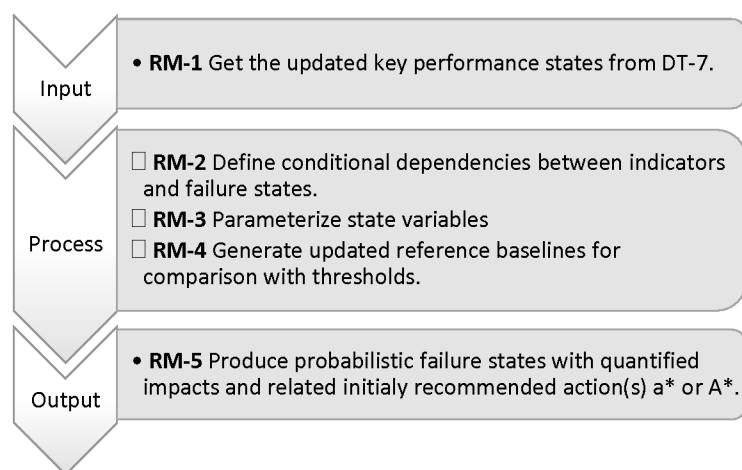
Figure 1—Flow of decision logic linking condition indicators to maintenance actions



**Figure 2—Data Assimilator Process – Multi-Source Data Acquisition and Preprocessing**



**Figure 3—Degradation Modeller Process: State Tracking & Forecasting**



**Figure 4—Risk Manager Process: Risk-State Inference from Indicators**

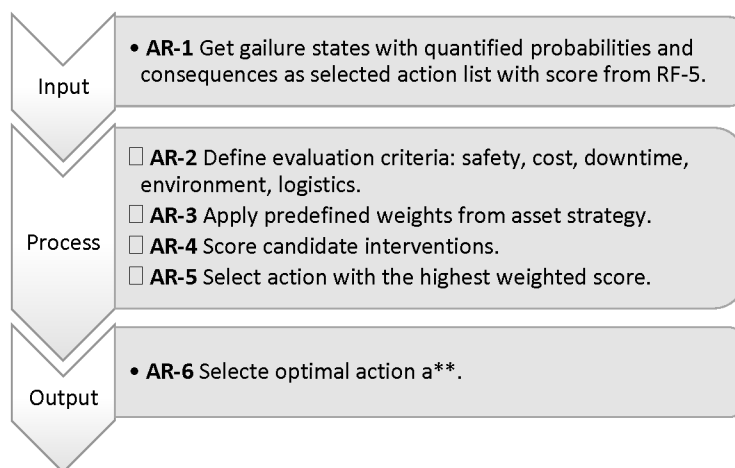


Figure 5—Action Recommender Process: Option Scoring and Playbook Selection

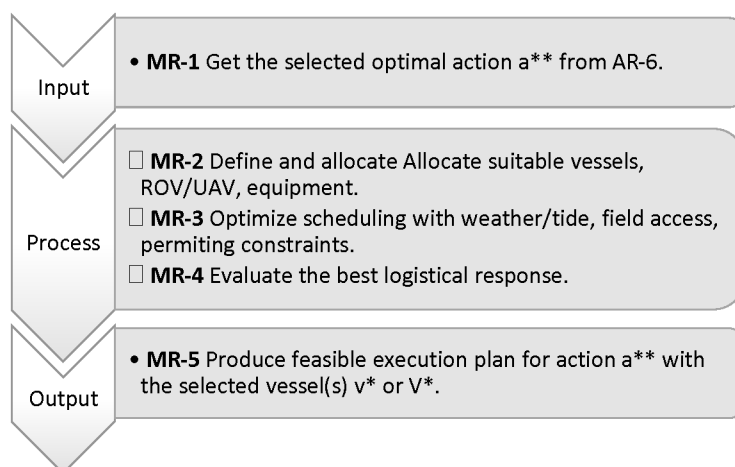


Figure 6—Marine Responder Process: Execution Plan and Vessel/Window Optimization

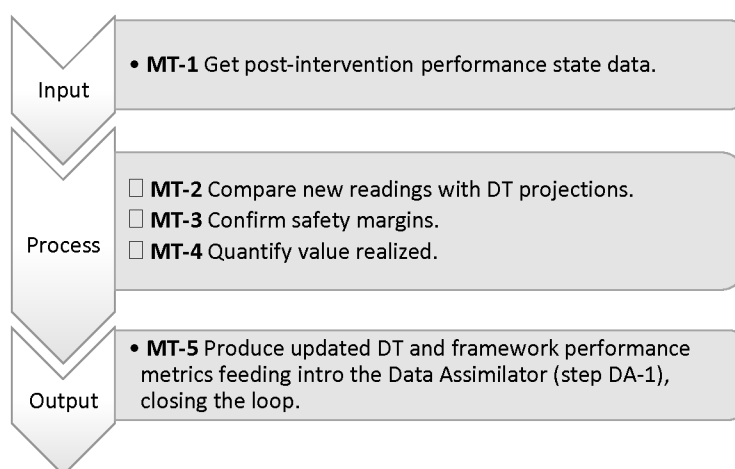


Figure 7—Model Tuner Process: Verification, Feeding and Governance Loop

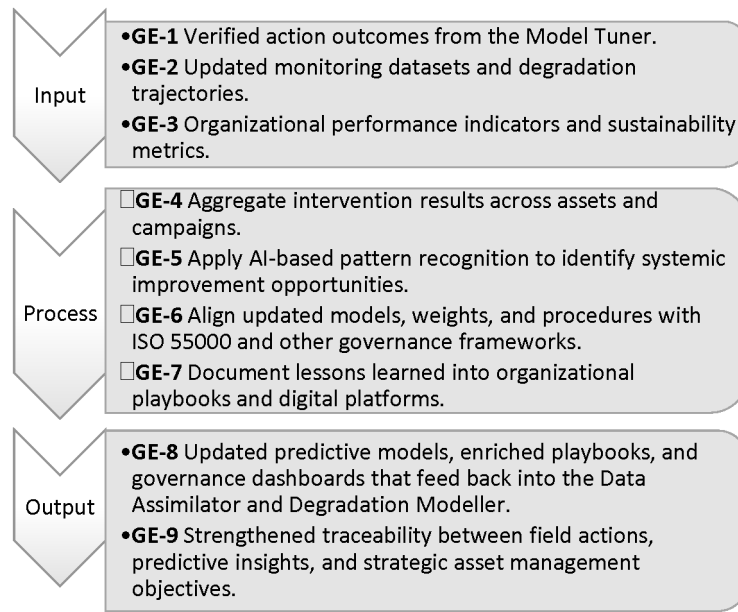


Figure 8—Governance Engine Process

The core of the BDN approach is Bayes' theorem, which allows updating the belief about the likelihood of a failure state  $F$  given an observed indicator  $I$ :

$$\text{posterior risk of failure} = \frac{\text{Likelihood of observing indicator given failure} \times \text{Prior risk of failure}}{\text{Total probability of the indicator}}$$

Mathematically, this is expressed as:

$$P(F|I) = \frac{P(I|F)P(F)}{P(I)} \quad \text{Eq. 3}$$

Or also, for each failure state  $f_k$ :

$$P(f_k|I) = \frac{P(I|f_k) \times P(f_k)}{\sum_{j=1}^4 P(I|f_j)P(f_j)} \quad (k = 1 \text{ to } 4) \quad \text{Eq. 4}$$

Where:

$F$  = the set of possible failure states  $F = \{\text{Stable, Early Warning, Critical Warning, Imminent Failure}\}$

$I$  = the set of possible indicators, e.g.,  $I = \{\text{settlement rate, UT wall thickness, CP potential, strain rate}\}$

$P(F|I)$  is the posterior probability (updated belief)

$P(I|F)$  is the likelihood of observing the indicator if failure exists

$P(F)$  is the prior probability of failure

$P(I)$  is the marginal probability of observing the indicator

Then, the utility or decision node, a numerical scope that reflects how desirable or undesirable an outcome is in the BDN, associated with the digital asset, selects the optimal action  $a^*$ , from all actions  $a$ , which belongs to the predefined group of actions  $A$ , that maximizes the expected utility  $EU$ , which is expressed as follows:

$$EU(a) = \sum_{k=1}^4 U(a, f_k) \times P(f_k|I) \quad \text{Eq. 5}$$

Where:

$A$  = the set of possible actions, e.g.,  $A = \{\text{Monitor more frequently, Partial Intervention, Full intervention}\}$

$U(a,f)$  is the utility of taking action  $a$  if state  $f$  is true, in another term, it is how good or bad action  $a$  is if the true state is  $f$ .  $U(a,f)$  is normalized on  $[-1,1]$ .

$P(f|I)$  is the posterior probability if state  $f$  is true given  $I$ , in another term, how likely you think  $f$  is, given the evidence  $I$ .

The sum would then give a weighted average outcome of taking the set action  $a$ . Then, to get the optimal action  $a^*$ , we run the above Eq. 5, for each possible action  $a$  that belongs to  $A$ , and we compute the expected utility  $EU$ , then we select the action that gives the highest value. It can be expressed as follows:

$$a^* = \underset{a \in A}{\operatorname{argmax}} EU(a) \quad \text{Eq. 6}$$

The outcome of this key decision layer is in fact a theoretical optimal action based on risk-weighted value. However, there is a need to review and select the safe, most practical, timely, and cost-effective action in case of obtained similar expected utility with different actions e.g.,  $a_1$ ,  $a_2 (=a^*)$  and  $a_3$ . with a respective expected utility of  $EU(a_1)=0.50$ ,  $EU(a_2=a^*)=0.51$  and  $EU(a_3)=0.49$ . In this case with introduce a multi-criteria decision making, see §2.2.4, as part of the Recommender stage. A practical example of the use of the BDN-driven framework is provided in §4.4

**Multi-Criteria Decision Making.** In practice, the probabilistic modelling might result in several actions with comparable utility values or unviable logistical actions. Therefore, we introduced an additional multi-criteria decision-making layer to refine the choice based on multiple real-world constraints. The MCDM method consists in assigning weighted scores to each candidate across the following typical key criteria:

- Cost
- Safety
- Downtime
- Environmental impact
- Logistics complexity

The most common MCDM techniques include weighted-sum models. In this work, we use a normalized weighted scoring model with (assumed) stakeholder defined weights as follows:

$$S(a_j) = \sum_{i=1}^I w_i \times r_{i,j} \quad \text{Eq. 7}$$

Where:

$S(a_j)$ = the score per action

$A^{**}$ = the candidate/selected actions set  $\{a_j, \dots, a_j\}$  from the BDN

$C$ = the MCDM criteria set  $\{c_1, \dots, c_I\}$

$w_i$ = the criterion weight, where  $w_i \geq 0$  and  $\sum_i w_i = 1$

$r_{i,j}$ = the normalized score of action  $a_j$  on criterion  $c_i$ , where  $r_{i,j} \in [0,1]$ . If using 1–5 scoring is used

The outcome blends several practical criteria into a score per action. Then, the best action can be obtained by:

*Best action  $a^{**}$  = the option  $a$  with the highest (weight  $\times$  criterion score) across the MCDM criteria*

Mathematically, the relation becomes:

$$a^{**} = \underset{a \in A^{**}}{\operatorname{argmax}} S(a_j) \quad \text{Eq. 8}$$

The MCDM scoring matrix ranks viable options shortlisted by the BDN and provides a practical basis for selecting a playbook action  $a(**)$ , which corresponds to the highest  $S(a_j)$ . A practical example is presented in §0

**Marine Logistics Responder.** The role of the proposed marine logistics responder (MR) is to turn the recommended action (i.e.,  $a(**)$  in this work) from §2.8 into an executable plan under weather, tide/permit and resource constraints. The MR aims at choosing a compatible vessel and feasible start time to execute  $a(**)$  safely with optimal cost, time and carbon dioxide emission, contributing to the need for a shift toward sustainable marine logistics operations.

To translate the proposed solution into an optimization problem that can be solved analytically, the marine responder we are proposing will be based on the following equation:

$$\min J = \text{monetary cost of the vessel} + \text{makespan} + \text{CO}_2 \text{ emissions}$$

Mathematically, this can be expressed as:

$$\min_{v \in V} J(v) = w_c c_v + w_t T_v + w_e EF_v (\tau_v^{in} + d_v + \tau_v^{out}) \quad \text{Eq. 9}$$

Where:

$V$  = the set of candidate vessels.

$v$  = the individual vessel from the set  $V$

$w_i$  = the scalar weights for cost, makespan (duration), and emissions.

$c_v$  = the monetary cost of vessel  $v$  (e.g., charter rate/day, mobilization).

$T_v$  = the total operation time if vessel  $v$  is chosen (mobilization, transit, onsite duration).

$EF_v$  = the emission factor of vessel  $v$  (e.g., kg  $\text{CO}_2$  per hour).

$\tau_v^{in}$  = the inbound sailing duration.

$d_v$  = the duration of onsite work.

$\tau_v^{out}$  = the outbound sailing duration.

The optimizer's role is to simply find and select the minimum between all calculated  $J(v)$  for the vessels  $v$  part of the relevant set of vessels  $V$  for this operation, at the location  $l$ :

$$J^* = \min_{v \in V} J(v) \quad \text{Eq. 10}$$

And accordingly, the selected vessel  $v^*$  is where:

$$v^* = \arg \min_{v \in V} J(v) \quad \text{Eq. 11}$$

## Weights Across the Framework

The individual components of the framework, i.e., digital twin, degradation models, Bayesian networks, multi-criteria decision making, and the marine responder, become most effective when integrated into a consistent governance structure. Integration ensures that technical insights derived from monitoring and modeling can be translated into simple and clear actions. Applying weights contributes to creating a coherent decision-making process. They are applied at different layers of the process to ensure that decisions reflect both technical priorities and organizational objectives:

- **Design and Standards:** Weighting factors are informed by international codes such as BS 6349 and Eurocode.
- **Probabilistic Confidence:** In the Bayesian decision network, weights express the confidence in monitoring indicators and degradation models, dynamically updated as new data becomes available.
- **Decision Criteria:** Within the multi-criteria decision framework, weights allow asset owners to balance cost, safety, downtime, environmental impact, and logistics according to strategic priorities.

- **Logistics Optimization:** The marine responder applies weights to vessel availability, scheduling constraints, and emissions, optimizing intervention timing in terms of both operational risk and sustainability.

The weighting mechanisms connect design standards, probabilistic reasoning, governance preferences, and operational logistics into one consistent decision chain. It is worth mentioning, that the human capital is in fact the key contributor in setting appropriate weights aligning with the asset owner's strategic asset management plan and considering subject matter expertise.

## Procedures

This section describes the step-by-step procedure for implementing the proposed AI-enabled data-to-action framework. The procedures translate the theoretical foundations and definitions established earlier into a practical workflow, starting from data acquisition and continuing through predictive modeling, probabilistic updating, decision support, and logistics optimization.

### Data Assimilator

The Data Assimilator (DA) forms the first stage of the framework, ensuring that all monitoring evidence is systematically captured, processed, and aligned prior to predictive analysis. It encompasses the collection of raw signals, their preprocessing into consistent formats, and their assimilation into the digital twin environment. At this stage, AI-enabled anomaly detection and signal enhancement can be employed to filter noise, identify outliers, and prioritize critical signals, ensuring that the subsequent predictive analyses are driven by high-quality, reliable data.

This step establishes the foundation for all subsequent analyses by creating a unified, quality dataset. By leveraging AI for anomaly detection and data fusion, the framework not only improves efficiency in preprocessing but also enhances confidence in the integrity of the monitoring evidence.

### Degradation Modeller

The Degradation Modeller (DM) serves as the predictive anchor of the framework, combining the design basis with observed performance structured via the Data Assimilator. At this stage, the original creation of the digital twin is assumed, and the focus is on maintaining predictive capability through ongoing state updates.

Artificial intelligence contributes to enhancing the DM in (at least) two critical ways. First, an AI-based anomaly detection layer improves the reliability of monitoring data by flagging deviations or unusual patterns that may indicate early signs of deterioration. For instance, deep learning applied to InSAR time-series has been shown to detect abnormal ground settlement in coastal infrastructure before it becomes visible in traditional inspections (Zhang et al., 2022). Similarly, neural network models have been used to filter ultrasonic thickness measurements for marine piles, identifying hidden corrosion signatures.

Second, AI forecasting methods extend the ability of the DM to predict future states. Hybrid machine learning–physics models have been applied successfully to forecast long-term chloride-induced corrosion in marine steel structures with greater accuracy than deterministic models (Song et al., 2021). Recurrent neural networks (RNNs) and long short-term memory (LSTM) approaches have been employed to predict settlement and structural response from monitoring time-series (Liu et al., 2021), while convolutional neural networks (CNNs) have been tested for scour depth prediction around quay and bridge foundations (Wang et al., 2020). The proposed DM process is therefore structured as follows:

By embedding AI in anomaly detection and forecasting, the DM evolves from a purely mechanistic model into a dynamic predictive engine. This integration enables early recognition of deterioration trends, improves confidence in long-term forecasts, and supports proactive, risk-informed maintenance planning.

## **Risk Manager**

This stage implements the Bayesian decision network to translate DM state variables into probabilities of failure modes. It fuses indicators into risk states, creating a bridge between physical reality and decision-making.

The BDN transforms evidence into risk-informed states, forming the quantitative backbone for multi-criteria decision analysis. While urgency prioritization can be integrated in practice, this paper remains focused on the predictive risk values.

This approach can be embedded within the asset owner's organizational risk management methodology, specifically as part of the risk assessment process, aligning with ISO 31000 framework [7], where key metrics and criteria used in the assessment matrix of asset management systems, can be turned into thresholds against which, key failure state probabilities can be compared with.

## **Action Recommender**

In this step, the Action Recommender (AR) gets the risk figures and filters priority candidate interventions. Then, the selected actions undergo the multi-criteria decision making to assure that the selected action balances technical, economic, environmental, and operational aspects.

This step ensures that the chosen action reflects both risk mitigation and value optimization, preparing the right outputs for the marine responder.

## **Marine Responder**

The Marine Responder (MR) transforms a technical action into a real, feasible operation by orchestrating resources, assets, and windows of opportunity. It translates recommendations into mobilization.

The MR grounds the predictive framework in operational reality. At production scale, grouping actions across assets enhances efficiency and sustainability.

## **Model Tuner**

Verification ensures that the action taken achieves its intended outcome and feeds the results back into the learning loop.

This step closes the loop, embedding a governance cycle aligned with ISO 55000. Each intervention strengthens predictive accuracy for the future.

## **Governance Engine**

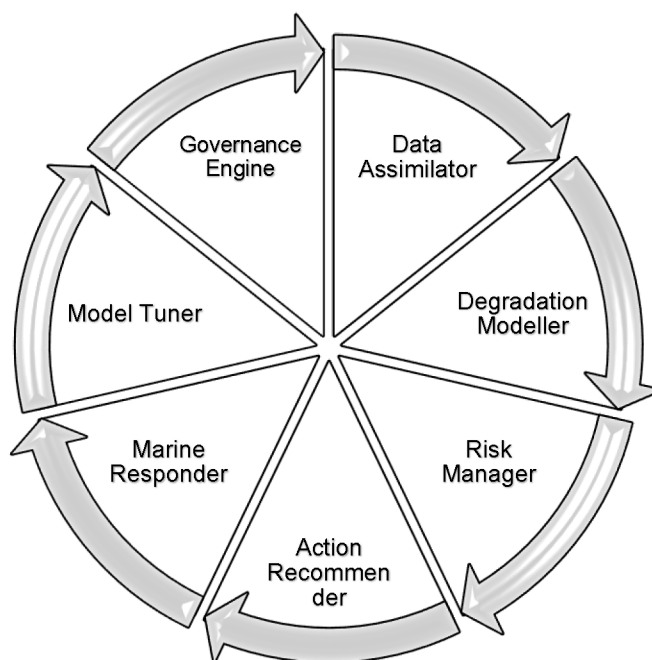
The Governance Engine (GE) provides the final integration step of the framework, ensuring that predictive insights, executed actions, and observed outcomes are consolidated into a continuous improvement cycle. It builds upon the outputs of the Model Tuner, embedding them into organizational asset management systems.

At this stage, AI plays a role in extracting lessons learned from both successful and unsuccessful interventions, identifying patterns across projects, and refining weighting schemes in the decision-support modules. By structuring this feedback, asset owners develop a living governance structure that not only maintains predictive capability but also strengthens accountability and transparency at strategic and operational levels.

By embedding governance and continuous learning, the framework evolves from a set of analytical tools into an organizational capability. This ensures that each monitoring cycle and intervention not only addresses immediate risks but also enhances resilience, efficiency, and sustainability at the system level.

## **Framework Core Process**

The proposed framework integrates seven engines (Fig. 9) into a continuous predictive loop. Each stage builds on the previous, transforming raw monitoring data into risk-informed, optimized, and governable maintenance actions.



**Figure 9—Proposed AI-Driven Predictive Analytics Core Process for Marine Civil Infrastructure**



**Figure 10—Escape-boat jetty on remote artificial islands**

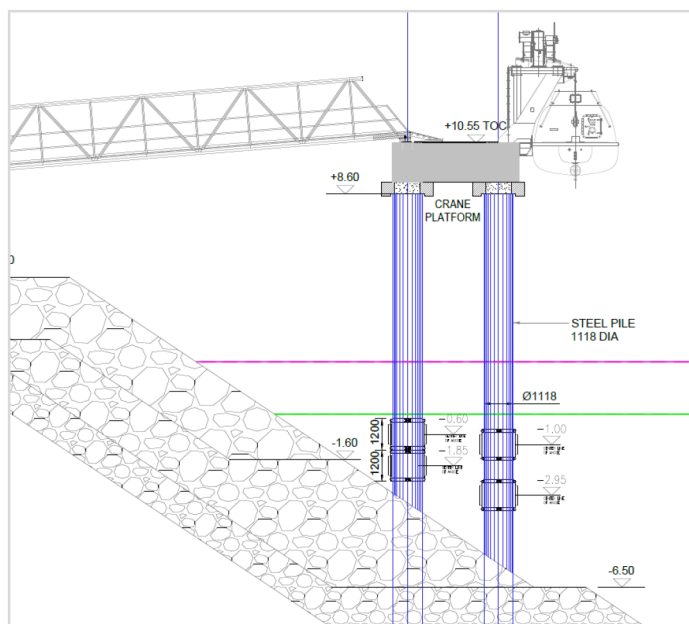


Figure 11—Cross section of the escape-boat jetty

The result is a transparent, AI-enabled process that enhances resilience, reduces lifecycle costs, and aligns marine civil infrastructure management with engineering and operations best practice.

## Simulation: Escape-boat jetty on Offshore Energy Islands

### Introduction

Escape-boat jetties on remote drilling and processing artificial islands represent a life-critical asset, providing a primary evacuation route in the event of high-risk emergencies such as  $H_2S$  releases during drilling or production operations. The integrity of their supporting steel tubular piles, particularly in the highly aggressive splash zone of the Arabian Gulf, is therefore central to both personnel safety and regulatory compliance.

While traditional design practice incorporates sacrificial wall thickness, coatings, and cathodic protection (CP) to extend service life, unexpected, localized corrosion or under-performing CP (generally in a form of sacrificial anodes) can still reduce residual thickness below safe limits well before design life is reached.

This section demonstrates, step by step, how adopting the proposed framework core processes, as suggested in this work in Fig. 9, provides a structured, evidence-based pathway to forecast risk, evaluate intervention strategies, and select the most effective action.

For the purposes of the following simulation, the digital twin or dedicated smart web-based platform is assumed to be already established and operational, serving as the backbone for data integration, state tracking, and decision support. This scope of this work does not address the development or deployment of the digital twin itself but rather demonstrates how an existing digital twin environment enables the predictive maintenance framework and supports the integration of degradation models, Bayesian decision networks, multi-criteria decision making, and marine logistics coordination.

### Step 1 – Data Assimilator

For this asset class, the structural design life is typically 50 years, while the cathodic protection (CP) system design life is typically 25 years, after which sacrificial anodes require replacement. Our objective intervention is therefore evaluated against two target performance indicators:

- Immediate safety: restore capacity so that the capacity margin now and across the operational planning horizon (e.g., 12 months).
- Remaining design life: ensure the corrosion trajectory with a working CP system keeps sufficient capacity margin to the end of the remaining design life (or to the next scheduled CP renewal).

The key indicators to be tracked in this simulation are:

- The steel pipe wall thickness  $th$
- Corrosion rate  $r$
- CP potential  $E_{cp}$
- Pile capacity margin  $M$

Recent ROV/UT inspections revealed the following indications:

- $th=15.8 \text{ mm}$ , where the safe design min is  $20 \text{ mm}$  or more, approaching the critical threshold is  $14 \text{ mm}$ .
- $r=0.35\text{mm/yr}$ , as observed from the business-as-usual inspections.
- $E_{cp}= -700 \text{ mV}$  with target to reach below  $-850 \text{ mV}$  which corresponds to the maximum potential to keep the steel surface polarized sufficiently to suppress anodic dissolution when considering an Ag/AgCl reference electrode.
- $M=1.05$ , capacity margin close to unity, under the most critical design load, suggested limited margin.

## Step 2 – Degradation Modeller

The degradation process is projected using a linear corrosion-loss model, where the residual thickness decreases proportionally with the measured corrosion rate  $r$ .

$$th(t + \Delta t) = th(t) - r\Delta t \quad \text{Eq. 12}$$

These forecasts are passed directly to the BDN, which interprets them probabilistically as failure-state beliefs. No intermediate classification is required; the DM supplies physics-based trajectories, and the BDN converts evidence into updated state probabilities.

## Step 3 - Risk Manager

Let set  $F=\{f_1, f_2, f_3, f_4\}$  denote the failure states, linked to the performance grades presented earlier on in §2.1.6, defined as follows:

- $f_1 \equiv$  Stable: Residual thickness  $\geq 20 \text{ mm}$ , i.e., design intent met, ample capacity margin, business as usual.
- $f_2 \equiv$  Early Warning:  $16\text{--}20 \text{ mm}$ , i.e., below allowance, requires closer monitoring, CP reinforcement.
- $f_3 \equiv$  Critical Warning:  $14\text{--}16 \text{ mm}$ , i.e., approaching threshold, pre-authorized playbooks triggered, localized repair or strengthening needed.
- $f_4 \equiv$  Imminent Failure:  $< 14 \text{ mm}$ , i.e., capacity compromised, failure likely, urgent intervention required.

The indicator set is defined in this case as  $I=\{th, r, E_{cp}, M\}$ . The posterior probability (belief) will be:

- $P(f_1 | I)=0.20$

- $P(f_2 | I) = 0.35$
- $P(f_3 | I) = 0.30$
- $P(f_4 | I) = 0.15$

The action set is defined as  $A = \{a_1, a_2, a_3, a_4\}$  defined as follows:

- $a_1$ : Intensify monitoring
- $a_2$ : Replace anodes
- $a_3$ : Clamp-on composite sleeve at splash zone + anodes replacement
- $a_4$ : Full pile replacement

The expected utility per action and state failure, from -1 to +1, were higher means better outcome, can be set as:

Hence, BDN decision is  $a^* = a_4$ , which means a recommendation to carry a full pile replacement.

However, by comparing the outcome in Table 2, we can see that the BDN selected action  $a^* = a_4$  has a comparable expected utility  $EU$  as to action  $a_3$ , which is the partial intervention of clamping-on composite sleeve to strengthen the section (e.g., over the splash zone) and replacing depleted sacrificial anodes, to extend the life of the CP system. This justify the need to apply a further layer of decision making with clear strategic criteria, i.e., MCDM in this work, which takes into account cost, downtime, carbon dioxide emissions, to finally chose the most practical and optimal action.

Table 2—Risk Manager's expected utility per action and state failure

| Action\State | $f_1$ | $f_2$ | $f_3$ | $f_4$ | Expected Utility  |
|--------------|-------|-------|-------|-------|-------------------|
| $a_1$        | +0.9  | -0.4  | -0.9  | -1.0  | $EU(a_1) = -0.38$ |
| $a_2$        | +0.6  | +0.1  | -0.5  | -0.8  | $EU(a_2) = -0.11$ |
| $a_3$        | +0.3  | +0.7  | +0.9  | +0.6  | $EU(a_3) = +0.66$ |
| $a_4$        | -0.2  | +0.8  | +1.0  | +1.0  | $EU(a_4) = +0.69$ |

Table 3—Action Recommender's MCDM weighted-sum score

| Action\Criterion | Cost | Safety | Downtime | Envrinmental | Logistics | Weighted Score  |
|------------------|------|--------|----------|--------------|-----------|-----------------|
| $a_3$            | 6    | 9      | 6        | 8            | 7         | $S(a_3) = 7.45$ |
| $a_4$            | 3    | 10     | 3        | 6            | 4         | $S(a_4) = 6.00$ |

#### Step 4 – Action Recommender

We will calculate the MCDM weighted-sum score, assuming the following weights:

- Cost:  $w_1 = 0.25$
- Safety:  $w_2 = 0.35$
- Downtime:  $w_3 = 0.15$
- Environmental:  $w_4 = 0.15$
- Logistics:  $w_5 = 0.10$

Accordingly, the MCDM recommendation is  $a^{**}=a_3$ , confirming the BDN risk-based decision. Full replacement ensures safety but is infeasible within the 5-month risk horizon and is of a higher cost and results in a significant downtime of the escape boat jetty, until completion of the full pile replacement works.

### Step 5 - Marine Responder

The objective from this step is to define the optimal logistical response. The available logistical options, i.e., candidate vessels  $V=\{v_1, v_2\}$  are as follow:

- Vessel  $v_1$ : Workboat + dive team + clamp sleeve with a cost of  $c_1 = \$35k$ ; 24 h campaign, resulting in a total estimate emission of 20 t  $CO_2$ .
- Vessel  $v_2$ : Jack-up barge + crane, needed for replacement, with a cost of  $c_2 = \$180k$ ; 96 h campaign, resulting in a total estimate emission of 295 t  $CO_2$ .

Based upon the weights defined in the SAMP, we extract the following key weights for cost, time and carbon emissions:

- Cost:  $w_c=0.40$
- Time:  $w_t=0.40$
- Emission:  $w_e=0.2$  (depends on the sensivity of the location, e.g., Marine Protected Areas)

The objective function, as per §2.2.5 is defined as:

$$\min_{v \in V} J(v) = w_c c_v + w_t T_v + w_e EF_v \quad \text{Eq. 13}$$

After normalization, the result is  $J(v_1)=0.1$  and  $J(v_2)=1.0$ , therefore the  $v^*=v_1$  wins decisively, provided it is far cheaper, much faster, and it emits far less  $CO_2$ .

### Step 6 – Model Tuner

The final stage of the predictive maintenance cycle is the verification and valuation of outcomes. This step closes the loop between intervention and model, ensuring that both the engineering repair and the digital representation of the asset are consistent. It also provides the governance evidence necessary for asset managers and senior leadership to quantify the value created by proactive measures.

As part of this step, we verify and confirm the outcome of the action:

- Two weeks post-works: ultrasonic testing confirms effective restored wall thickness above 25 mm at the sleeve, and cathodic protection potentials remain below -850 mV (Ag/AgCl reference electrode), satisfying the recommended CP potential criteria.
- Digital Twin re-assimilation: updated information reduces the probability of critical wall loss  $P(f_3 \# I_{new})$  to less than 0.05, confirming the effectiveness of the action in the Bayesian decision network.

Then we proceed to the valuation step, where we consider key governance metrics:

- Safety margin restored, ensuring escape-jetty operability under an  $H_2 S$  emergency scenario.
- Avoided shutdown costs estimated at \$120–400k (e.g., providing a temporary replacement in case of emergency during the works and shutdown of the escape boat jetty, or halting drilling/production) compared with a full replacement campaign.
- Approx. 276 t  $CO_2$  emissions avoided through optimized vessel selection and targeted intervention.

By combining technical validation, digital re-assimilation, and financial valuation, this final step demonstrates the closed-loop governance of the framework. It not only secures the integrity of the asset but

also documents the tangible economic and sustainability gains, providing a strong case for scaling predictive approaches across the artificial island portfolio.

### Step 7 – Governance Engine

The Governance Engine is the final step that turns the technical results of the simulation into organizational value. In the jetty case, the verified improvements, i.e., restored wall thickness, stabilized corrosion potentials, and reduced probability of failure are recorded into the operator's asset management system. This creates a clear audit trail showing what data was collected, what actions were taken, and what outcomes were achieved. It gives senior managers confidence that interventions are not only technically sound but also cost-efficient and aligned with wider risk and sustainability targets.

The governance layer captures lessons learned and feeds them back into decision rules for future projects. By doing so, it ensures that successful strategies, such as combining clamps with improved cathodic protection, can be repeated and scaled across other assets.

## Result

The simulation of the escape-boat jetty demonstrated the practical value of the proposed data-to-action framework. By integrating monitoring evidence into the Degradation Modeller and Risk Manager, the probability of reaching a critical wall loss state was reduced by more than half once the clamp and cathodic protection intervention was applied.

The Action Recommender and Marine Responder translated this technical solution into an optimized maintenance campaign, lowering vessel mobilizations and associated CO<sub>2</sub> emissions by a significant percentage compared with a conventional reactive approach. Cost savings were achieved through an optimal preventive intervention, avoidance of costly reactive interventions and improving the asset resilience.

Finally, the Governance Engine would provide traceability of decisions and captured lessons learned, embedding them into organizational lessons learned for replication across other quay walls and revetments. Together, these results show how the framework converts complex monitoring data into transparent, risk-informed, and resource-efficient actions that strengthen resilience and sustainability of remote energy island infrastructure. In addition, the GE would allow asset managers to plan grouping of similar actions, which further compresses cost and vessel days by bundling inspections or light repairs within shared weather/permit windows.

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## Nomenclature

|      |                           |
|------|---------------------------|
| AI:  | Artificial Intelligence   |
| AR:  | Action Recommender        |
| BDN: | Bayesian Decision Network |
| CP:  | Cathodic Protection       |
| DA:  | Data Assimilator          |
| DM:  | Degradation Modeller      |
| DT:  | Digital Twin              |

GE: Governance Engine  
InSAR: Interferometric Synthetic Aperture Radar  
ISO: International Organization for Standardization  
LSTM: Long Short-Term Memory (neural network model)  
MCDM: Multi-Criteria Decision Making  
ML: Machine Learning  
MR: Marine Responder  
MT: Model Tuner  
OECD: Organisation for Economic Co-operation and Development  
PIANC: Permanent International Association of Navigation Congresses  
RNN: Recurrent Neural Network  
RM: Risk Manager  
SDG: Sustainable Development Goal  
DL: Damage Limitation (Eurocode limit state)  
SD: Significant Damage (Eurocode limit state)

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